

QUESTION BANK

Unit – I [MATRIX ALGEBRA]

Basic concepts of Matrices and Determinants, Symmetric, Skew-symmetric matrices, Orthogonal matrices, Complex matrices, Hermitian and Skew-Hermitian matrices, Unitary matrices, triangular and diagonal matrices, Rank of a Matrix, Linear systems of equations – Existence and uniqueness of solutions, Gauss Elimination, Cramer's Rule, Matrix Inversion Method, Gauss Jordan elimination, LU factorization of matrices

LONG QUESTIONS

1.	If $C = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & -3 & 5 \end{bmatrix}$ then find $C^3 + C^2 + C$.
2.	Evaluate the following determinants by reducing to triangular form. $(i) \begin{vmatrix} 1 & 0 & 3 & 7 \\ 4 & 2 & 0 & 1 \\ 7 & 7 & 3 & 0 \\ 5 & 0 & 6 & 8 \end{vmatrix}$ $(ii) \begin{vmatrix} 2 & 0 & -4 & 6 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{vmatrix}$ $(iii) \begin{vmatrix} 3 & 1 & -1 & 0 \\ 2 & 2 & 2 & 1 \\ -1 & 3 & 0 & 4 \\ 8 & 6 & -2 & 2 \end{vmatrix}$
3.	Find the rank of the following matrices. $(i) \begin{bmatrix} 9 & 3 & 1 & 0 \\ 3 & 0 & 1 & -6 \\ 1 & 1 & 1 & 1 \\ 0 & -6 & 1 & 9 \end{bmatrix}$ $(ii) \begin{bmatrix} 3 & -1 & 5 \\ 2 & -4 & 6 \\ 10 & 0 & 14 \end{bmatrix}$ $(iii) \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$ $(iv) A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 5 & 8 \\ -3 & 4 & 4 \\ 1 & 2 & 4 \end{bmatrix}$
4.	Find the rank of the following matrices after reducing them to row reduced echelon form. $(i) \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix}$ $(ii) \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ $(iii) \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$
5.	Find the rank of the following matrices using determinant. $(i) \begin{bmatrix} 21 & -3 & 17 & 13 \\ 46 & 11 & 52 & 14 \\ 33 & 48 & 71 & -23 \end{bmatrix}$ $(ii) \begin{bmatrix} 3 & -1 & 5 \\ 2 & -4 & 6 \\ 10 & 0 & 14 \end{bmatrix}$ $(iii) \begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$

6.	If $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 4 & 1 \\ 3 & 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 5 & -1 \\ 0 & 2 & -2 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 7 & 2 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{bmatrix}$ then find (i) $A^3 + B^2 + C$, (ii) $A^2 + B^T + C(2A^T - 3B)$
7.	Solve the following systems of equations by Gauss elimination method. (i) $-x+y+2z = 2$, $3x-y+z = 6$, $-x+3y+4z = 4$. (ii) $x+y-z=9$, $8y+6z = -6$, $-2x+4y-6z = 40$. (iii) $-x+2y-5z = -21$, $4x-8y+3z = 16$, $3x-6y+z = 7$. (iv) $2w-3x-3y+6z = 2$, $5x+5y-10z = 0$, $4w+x+y-2z = 4$, $w+z = 10$.
8.	Find the inverse of the following matrices by both Gauss– Jordan Elimination Method and using determinant. (i) $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 1 & 1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ (iii) $\begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ (iv) $\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$
9.	Find the inverse transformations of the following. (i) $y_1 = 5x_1 + 3x_2 - 3x_3$ $y_2 = 3x_1 + 2x_2 - 2x_3$ $y_3 = 2x_1 - x_2 + 2x_3$ (ii) $y_1 = 0.2x_1 - 0.1x_2$ $y_2 = -0.2x_2 + 0.1x_3$ $y_3 = 0.1x_1 + 0.1x_3$
10.	Test the consistency of the following systems of equations and incase of consistent system conclude whether they have Unique solution, infinitely many solutions. (i) $2x-7y+4z=9, x+9y-6z=1, -3x+8y+5z=6$ (ii) $x - 2y + z = 4$, $3x + y - 2z = -7$, $x - 4y + 2z = 6$. (iii) $x+y-z=0$, $2x-y+z=3$, $4x+2y-2z=2$. (iv) $7x + 16y - 7z = 4$, $2x + 5y - 3z = -3$, $x + y + 2z = 4$.
11.	Solve the following system of linear equations using Cramer's rule. (i) $x+2y+3z=20, 7x+3y+z=13, x+6y+2z=0$. (ii) $3x-y+z=11, x+y+3z=13, 2x-3y+4z=23$. (iii) $3x+7y+8z=-13, 2x+9z=-5, -4x+y-26z=2$. (iv) $x+y=3, 2x-y+2z=-2, 3x+y=3$.

12.	Determine the values of a and b for which the following systems of linear equations are (i) consistent with unique solution, (ii) consistent with infinitely many solutions, (iii) inconsistent. (a) $x+2y+3z=6$, $x+3y+5z=9$, $2x+5y+az=b$ (b) $2x+3y+5z=9$, $7x+3y-2z=8$, $2x+3y+az=b$
13.	Solve the following systems of equations by Matrix Inversion method. (i) $x+y+z=3$, $x+2y+3z=4$, $x+4y+9z=6$. (ii) $x+2y+3z=1$, $3x-y+z=2$, $4x+2y+z=3$. (iii) $5x+3y+3z=48$, $2x+6y-3z=18$, $8x-3y+2z=21$. (iv) $x+y+z=8$, $x-y+2z=6$, $9x+5y-7z=44$.
14.	Solve the following systems of equations by Gauss Jordan Elimination method. (i) $2x+y+2z+w=6$, $6x-6y+6z+12w=36$, $4x+3y+3z-3w=-1$, $2x+2y-z+w=10$. (ii) $2x+y+4z=11$, $x+5y+7z=15$, $3x+11y+13z=25$. (iii) $x-y+2z=3$, $x+2y+3z=5$, $3x-4y-5z=-13$. (iv) $2w-x+y-z=-10$, $2w+3x-y-z=0$, $w+x+y-z=20$, $3w+x-y+z=60$.
15.	Compute LU- Decomposition/Factorization of the following matrices (i) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 6 & 9 & 8 \end{bmatrix}$ (iii) $\begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{bmatrix}$
16.	Compute LU- Decomposition/Factorization of the following symmetric matrices (i) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 0 & 6 \\ 0 & 5 & 7 \\ 6 & 7 & 8 \end{bmatrix}$ (iii) $\begin{bmatrix} 10 & 20 & 30 \\ 20 & 100 & 50 \\ 30 & 50 & 200 \end{bmatrix}$
17.	Solve the following systems of equations by LU- factorization method. (i) $x+y+2z=-1$, $4x+5y+13z=-7$, $10x+14y+43z=-25$. (ii) $3x+5y+2z=8$, $8y+2z=-7$, $3x+y+4z=13$. (iii) $4x+2y+14z=14$, $2x+17y-5z=-101$, $14x-5y+83z=155$. (iv) $5x+9y+2z=24$, $9x+4y+z=25$, $2x+y+z=11$.

SHORT QUESTIONS

1.	For $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 5 \\ 1 & -1 \end{bmatrix}$ show that $(A+B)^T = A^T + B^T$
2.	If $A = \begin{bmatrix} 2 & 1 \\ 1 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 5 \\ 0 & 8 \end{bmatrix}$ then find the value of $(B^T - A^T)^T$.
3.	If $C = \begin{bmatrix} 6 & 0 & 3 \\ 1 & 0 & -5 \end{bmatrix}$ and $D = \begin{bmatrix} 4 & 0 & -4 \\ -3 & 4 & 9 \end{bmatrix}$ then find the value of $6(C^T + 3D^T)^T$.
4.	If $A = \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -3 \\ 0 & 2 \\ 0 & 1 \end{bmatrix}$ then find the value of $A^T B$.
5.	Find $a \bullet b$ where $a = [5 \ 1 \ 2 \ -3]$ and $b = [1 \ 2 \ 3 \ 4]$.
6.	If $a = [3 \ 0 \ 4]$, $b = [-1 \ 8 \ 2]$, $c = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$, $d = \begin{bmatrix} 2 \\ -2 \\ 6 \end{bmatrix}$ then find $a - b + c^T - d^T$.
7.	For any two matrices A and B prove that $(A B)^T = B^T A^T$
8.	Find the minors and cofactors of the determinant $\begin{vmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{vmatrix}$
9.	Solve the system of linear equations $5x-3y=37, -2x+7y=-38$ using Cramers' rule.
10.	Find the rank of the matrix using determinant. $\begin{bmatrix} 4 & 3 \\ -8 & -6 \\ 16 & 12 \end{bmatrix}$
11.	Solve the system of linear equations $6x + 4y = 2, 3x - 5y = -34$ by Gauss elimination method.
12.	Find the rank of the matrix $A = \begin{bmatrix} 8 & -4 \\ -2 & 1 \\ 6 & -3 \end{bmatrix}$.
13.	Find the value of k for which the system: $2x-3y=4, 4x-ky=8$ has infinitely many solution.
14.	Find the rank of the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$
15.	Find the inverse transformation $y_1 = 3x_1 + 2x_2, y_2 = 4x_1 + x_2$

16.	Show that the inverse of a symmetric matrix is symmetric.
17.	Show that $(A^2)^{-1} = (A^{-1})^2$ for any nonsingular matrix A .
18.	Show that $(A^T)^{-1} = (A^{-1})^T$
19.	Show that $(A^{-1})^{-1} = A$
20.	Show that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$
21.	Evaluate the value of the determinant $\begin{vmatrix} a & a & a \\ a & b & b \\ a & b & c \end{vmatrix}$ by reducing it to a triangular matrix.
22.	Evaluate the value of the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 4 & 3 \\ 9 & 9 & 9 \end{vmatrix}$ by reducing it to a triangular matrix.
23.	Let $A = \begin{bmatrix} 4a & 2a & 3a \\ 6b & 5b & 7b \\ 8c & 4c & 8c \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. Determine the values of a, b, c such that $AB = 4B$
24.	Show that the determinant value of an orthogonal matrix is ± 1
25.	Show that the determinant value of a unitary matrix has absolute value 1.
26.	If A is any square matrix, then show that $\frac{A+A^T}{2}$ is symmetric and $\frac{A-A^T}{2}$ is skew-symmetric.
27.	Express the following square matrices as the sum of a symmetric matrix and a skew-symmetric matrix: $(i) \begin{bmatrix} 3 & -4 \\ 6 & 0 \end{bmatrix}$ $(ii) \begin{bmatrix} 4 & 2 & -3 \\ 1 & 3 & -6 \\ -5 & 0 & -7 \end{bmatrix}$ $(iii) \begin{bmatrix} 3 & -4 & -1 \\ 6 & 0 & -1 \\ -3 & 13 & -4 \end{bmatrix}$
28.	Show that any square matrix can be written as a sum of a Hermitian matrix and a skew-Hermitian matrix.
29.	Is the matrix $A = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2}i \\ \frac{\sqrt{3}}{2}i & \frac{1}{2} \end{bmatrix}$ unitary? Justify your answer.